

Executive Business Report

End-to-End Sales Forecasting & Demand Intelligence System

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Executive Summary

The business shows a positive long-term sales trajectory, but demand varies substantially from month to month, creating both stockout and overstock risk. Historical analysis shows that stronger demand tends to occur toward the later months of the year, while no single month is the top seller in every year. Three forecasting approaches were compared, and XGBoost produced the most accurate historical predictions, with an average percentage error of 12.78%. The analysis also identified unusual high- and low-sales weeks and grouped product sub-categories into four distinct demand segments. The central business implication is that inventory should not follow a one-size-fits-all policy.

Key Findings from EDA and Forecasting

Sales increased over the four-year period, with the underlying monthly trend rising from roughly 40,000 in the earlier period to above 60,000 toward the end. Peak months differed by year: September led in 2015, November in 2016, December in 2017, and November in 2018. Among SARIMA, Prophet and XGBoost, **XGBoost performed best**, achieving MAE of 8,582.53, RMSE of 10,760.89 and MAPE of 12.78%, compared with Prophet's 14.99% MAPE and SARIMA's 24.39%.

Three-Month Sales Forecast with Confidence Ranges

The statistical forecast provides a central estimate and a planning range that reflects uncertainty. These ranges are plausible planning bands, not guaranteed minimums or maximums.

Month	Expected Sales	Planning Range	Business Interpretation
Jan 2019	48,057	21,353 – 74,762	Moderate expected demand; uncertainty remains wide.
Feb 2019	34,177	7,473 – 60,882	Lower central estimate and greatest downside risk.
Mar 2019	68,799	42,095 – 95,504	Strongest expected month in the three-month outlook.

March deserves the strongest inventory readiness, while February should be managed more cautiously because its expected sales are lower and its uncertainty range is broad.

Top Three Sales Anomalies and Likely Causes

- 1. Week ending 22 March 2015 — sales of 37,703.67:** the largest high-sales anomaly detected. A plausible cause is a major bulk order, temporary demand surge, or promotional campaign.
- 2. Week ending 2 December 2018 — sales of 35,998.90:** likely associated with holiday-season demand, year-end promotions, or seasonal campaigns.
- 3. Week ending 18 November 2018 — sales of 30,572.45:** another major late-year spike that may reflect festive demand, promotional activity, or large customer orders. These causes are hypotheses because the dataset does not include promotional, inventory, holiday-event, or customer-order context.

Product Demand Segmentation and Stocking Strategy

Demand Segment	Products / Evidence	Recommended Stocking Strategy
High Value, High Volatility	Copiers and Machines. Avg. order value: 1,952.76; volatility: 5,552.16.	Maintain moderate safety stock, monitor frequently, and avoid tying excessive capital to unpredictable inventory.
Lower Volume, Stable Demand	Bookcases, Appliances, Envelopes, Furnishings, Fasteners, Art, Paper and Labels. Lowest cluster volatility: 907.22.	Use lean inventory and regular replenishment because demand is comparatively stable.
High Volume, Established Demand	Accessories, Binders, Chairs, Phones, Storage and Tables. Highest average total sales: 239,495.78.	Prioritize availability and dependable replenishment cycles to reduce stockout risk.
Rapidly Growing Demand	Supplies. Average YoY growth: 192.84%, although from a smaller sales base.	Increase stock gradually, review demand frequently, and avoid overreacting to percentage growth alone.

Three Concrete Business Recommendations

- 1. Protect availability in the highest-volume segment.** The High Volume, Established Demand cluster has average total sales of 239,495.78, the highest among all four segments. Accessories, Binders, Chairs, Phones, Storage and Tables should receive priority in replenishment schedules and stockout monitoring.
- 2. Prepare more inventory capacity for March while keeping February flexible.** The forecast expects approximately 68,799 in March versus 34,177 in February. Because forecast ranges remain wide, use phased purchasing and supplier flexibility rather than committing all inventory at once.
- 3. Use XGBoost as the primary short-term forecasting model and investigate extreme weeks.** XGBoost achieved the lowest historical error at 12.78% MAPE. Major anomalous weeks should trigger operational review for promotions, bulk orders, stock issues or seasonal events.

Business Risk / Limitation

The system learns only from historical sales and calendar patterns. It does not know about future promotions, price changes, stockouts, competitor actions, economic conditions, supplier disruptions or one-off bulk orders. Forecasts should therefore support—not replace—management judgment, and the models should be retrained as new data becomes available.